

RBI Repo Rate & CPI Inflation: Correlation Analysis (2016–2024)

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1. Objective

This project examines whether RBI repo rate changes influence retail inflation (CPI) in India, and whether any impact is immediate or delayed. Using monthly data from January 2016 to December 2024 (108 observations), the analysis combines SQL-based exploratory queries with Python-based Pearson correlation at multiple lag periods.

2. Data Sources

Variable	Source	Period
CPI Index & YOY Inflation %	FRED / OECD	Jan 2016 – Dec 2024
Repo Rate (%)	Reserve Bank of India	Jan 2016 – Dec 2024

3. Descriptive Statistics

Metric	Value
Total Observations	108 months (Jan 2016 – Dec 2024)
Average YOY Inflation	5.08%
Average Repo Rate	5.80%
Average CPI Index	125.8
Highest Inflation	9.63% (December 2019)
Lowest Inflation	1.08% (June 2017)
Highest Repo Rate	6.75% (2016)
Lowest Repo Rate	4.00% (2020–2021, Covid period)

4. Correlation Analysis

Lag Period	Pearson Correlation Coefficient
No Lag (Contemporaneous)	-0.1774
3-Month Lag	-0.1804
6-Month Lag	-0.1939
12-Month Lag (Strongest)	-0.2592

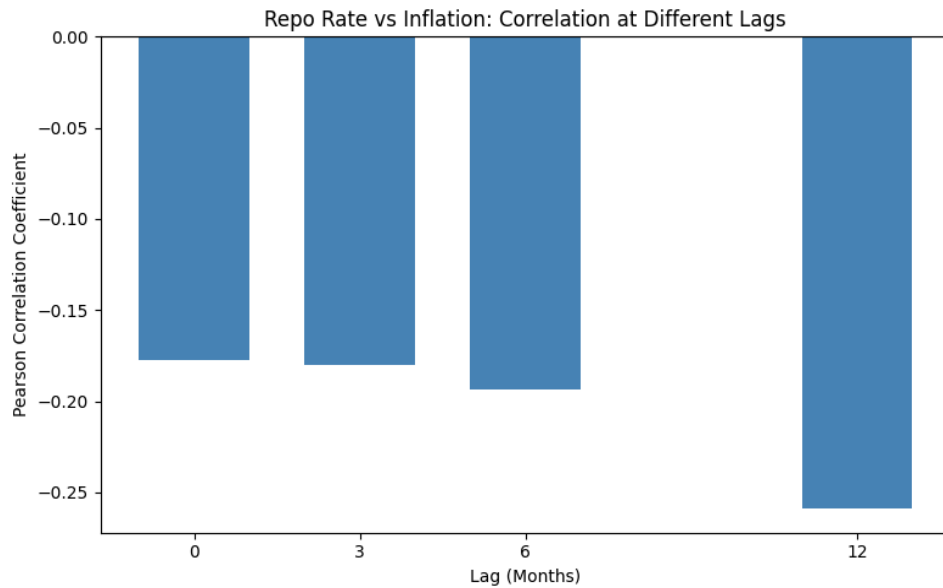


Figure 1: Pearson Correlation Coefficient at Varying Lag Periods

5. Key Findings

- Weak negative correlation overall: Repo rate and CPI inflation show a weak negative relationship across all lag periods, indicating that higher repo rates are associated with marginally lower inflation — but the effect is not strong.
- Lag effect confirmed: The correlation strengthens consistently from -0.18 (no lag) to -0.26 (12-month lag), confirming that monetary policy transmission to inflation takes approximately 12 months — consistent with RBI's own assessment of policy lag.
- 2019 anomaly: Inflation peaked at 9.63% in December 2019 while RBI was already cutting rates — a classic case of supply-side food price shocks that repo rate adjustments cannot directly address.
- Covid period (2020–21): Repo rate was cut to a historic low of 4.0%, yet inflation remained relatively contained at 4.89% average — driven by demand suppression rather than monetary policy effectiveness.
- Rate category finding (SQL): Months with Low repo rates (<5%) saw average inflation of 5.27%, versus 5.02% during High rate periods (>6.5%) — a counterintuitive finding explained by the lag: RBI cuts rates when inflation is already rising, so low-rate periods overlap with high-inflation episodes.

6. Conclusion

The analysis finds that RBI repo rate changes have a statistically weak but directionally consistent negative effect on CPI inflation in India, with the strongest impact observed at a 12-month lag ($r = -0.26$). The weak magnitude reflects that Indian inflation is significantly driven by food and fuel prices — supply-side factors that monetary policy cannot directly control. These findings align with established macroeconomic theory on monetary policy transmission lags and are consistent with RBI's documented policy stance of maintaining rates for sustained periods before inflationary effects materialize.

7. Tools & Techniques Used

Tool	Application
SQLite / DB Browser	Exploratory analysis — averages, distributions, year-wise trends, CASE bucketing, LEAD window function for lag
Python (pandas, scipy, matplotlib)	Pearson correlation at 0, 3, 6, 12-month lags; lag visualization chart

Tool	Application
Excel	Data cleaning and preparation